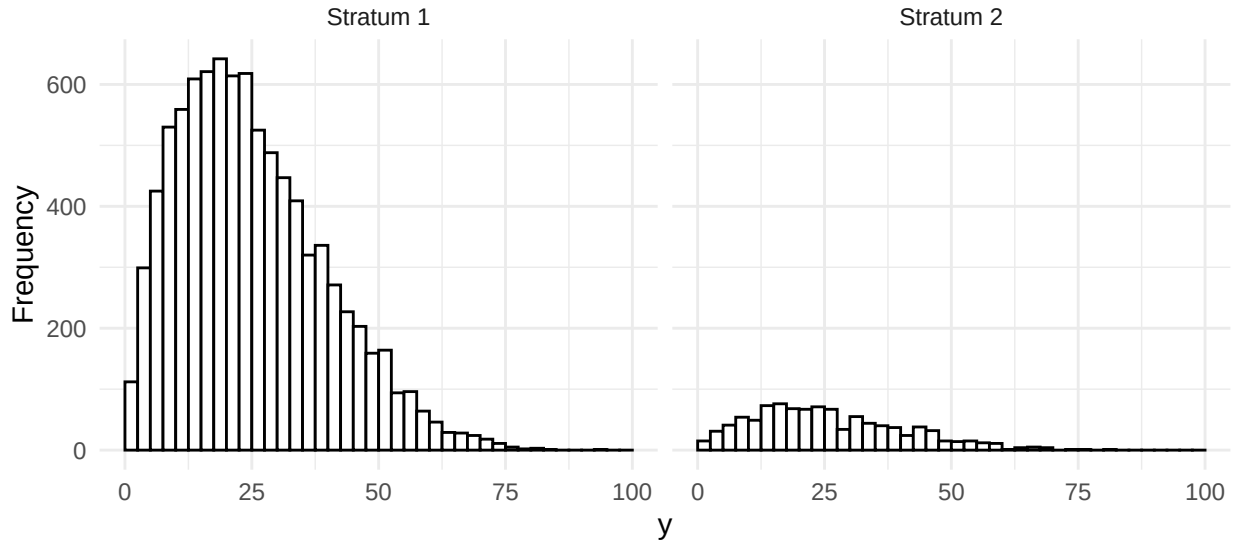
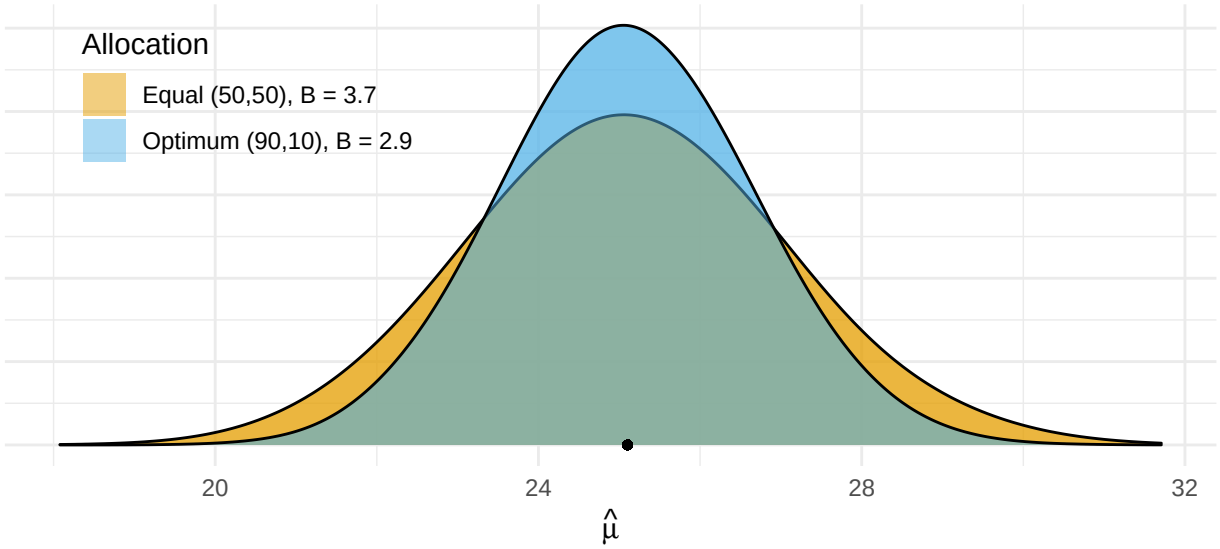
Monday, Sep 12

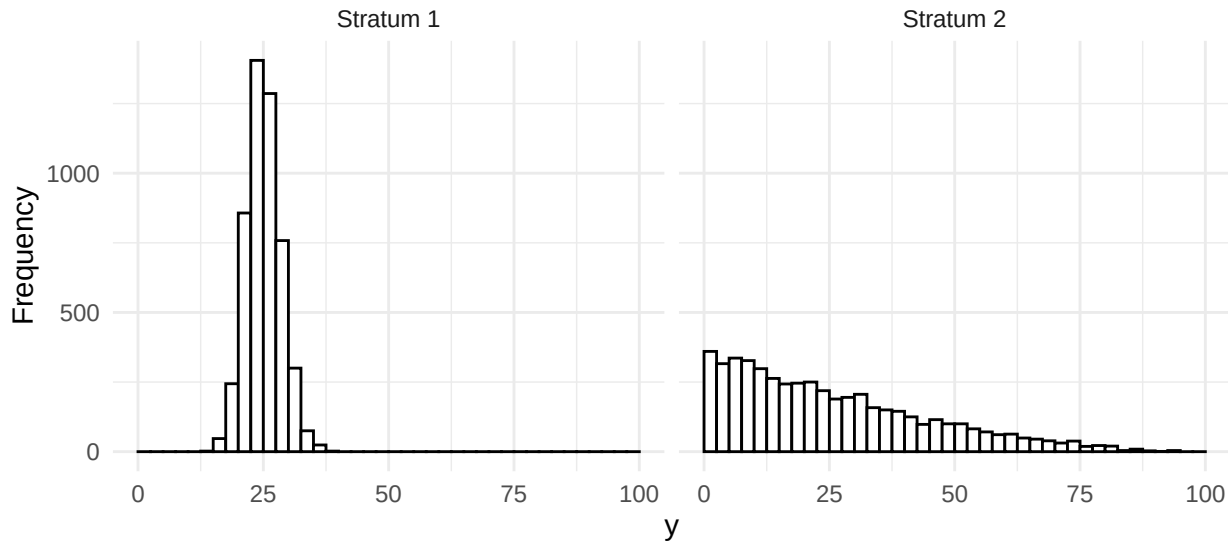
Simulation Study of Optimum Allocation

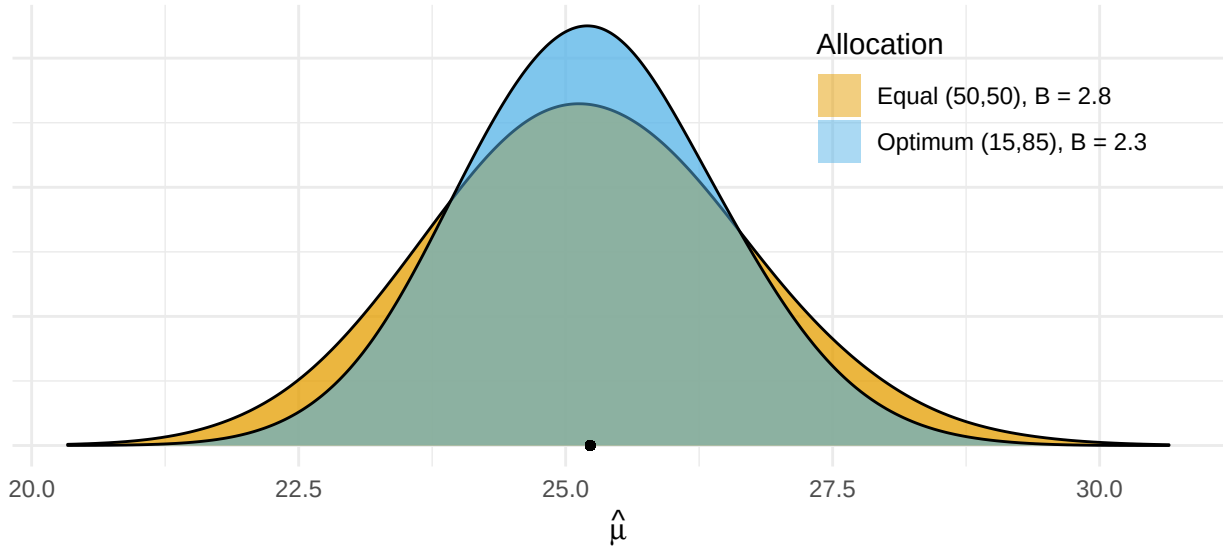
Strata have same variance, but different sizes.



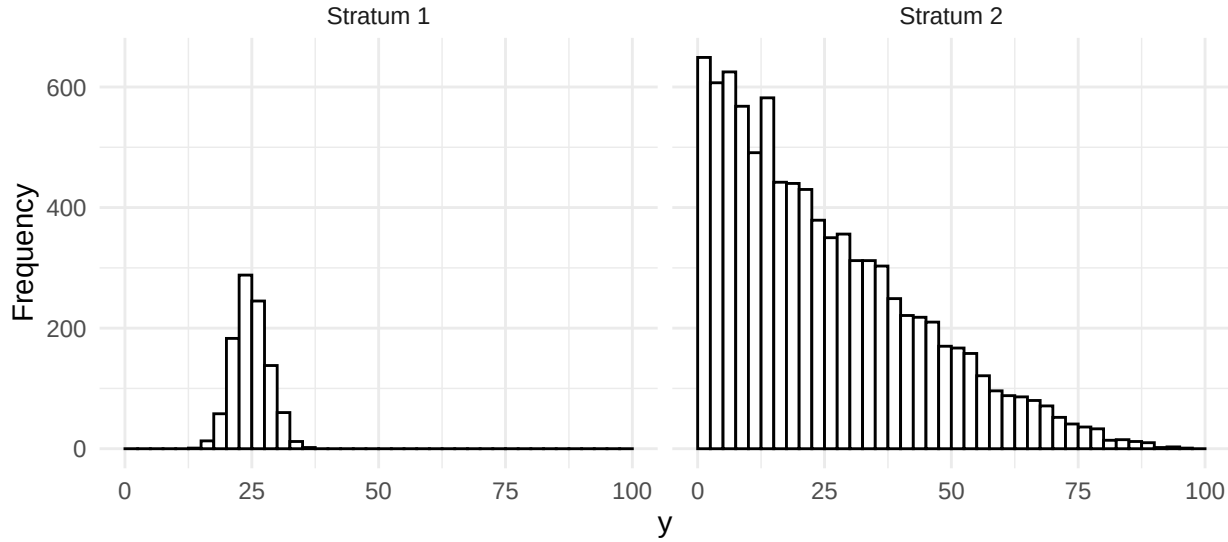


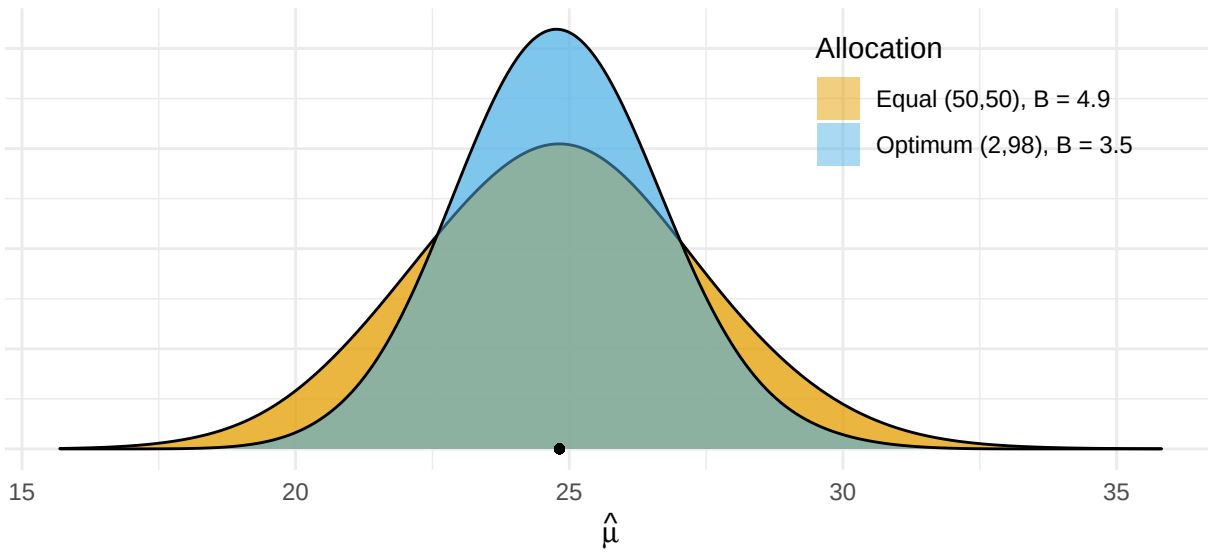
Strata have different variance, but same sizes.





Strata have different variances and sizes.





Comparison of Stratified and Simple Random Sampling

How does stratified random sampling compare with simple random sampling with respect to the variance of the estimators?

Analytical Results

Let V_{Opt} and V_{Prop} represent the variance of the estimator $\hat{\tau}$ or $\hat{\mu}$ from a stratified random sampling design with optimum and proportional allocation, respectively. And let V_{SRS} represent the variance of the estimator for a simple random sampling design. It can be shown that generally

$$V_{\text{SRS}} \geq V_{\text{Prop}} \geq V_{\text{Opt}}.$$

But *how* different are these variances?

1. The difference between *proportional* and *optimum* allocation stratified random sampling designs is

$$V_{\text{Prop}} - V_{\text{Opt}} = \frac{1}{n}\sum_{j=1}^{L}\frac{N_j}{N}(\sigma_j - \bar{\sigma})^2, \quad \text{where} \quad \bar{\sigma} = \sum_{j=1}^{L}\frac{N_j}{N}\sigma_j.$$

   So when is optimum allocation substantially better than proportional allocation in stratified random sampling?

2. The difference between *simple random sampling* and *proportional allocation stratified random sampling* is

$$V_{\text{SRS}} - V_{\text{Prop}} \approx \frac{1}{n}\left(1 - \frac{n}{N}\right)\sum_{j=1}^{L}\frac{N_j}{N}(\mu_j - \mu)^2.$$

   So when is proportionally-allocated stratified random sampling substantially better than simple random sampling?

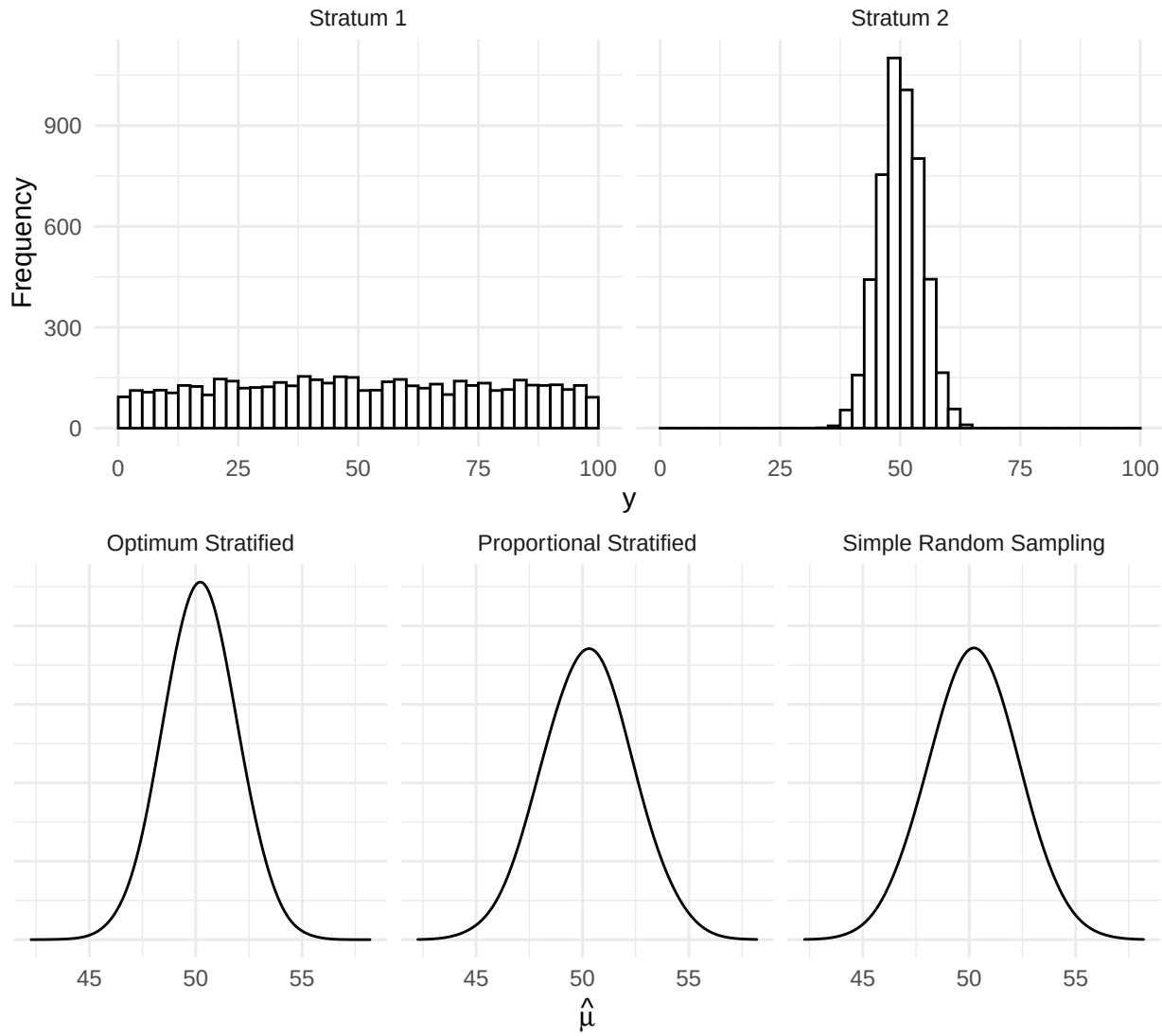
3. The difference between *simple random sampling* and *optimum-allocated stratified random sampling* is

$$V_{\text{SRS}} - V_{\text{Opt}} \approx \frac{1}{n}\sum_{j=1}^{L}\frac{N_j}{N}(\sigma_j - \bar{\sigma})^2 + \frac{1}{n}\left(1 - \frac{n}{N}\right)\sum_{j=1}^{L}\frac{N_j}{N}(\mu_j - \mu)^2.$$

So when is optimum-allocated stratified random sampling substantially better than simple random sampling?

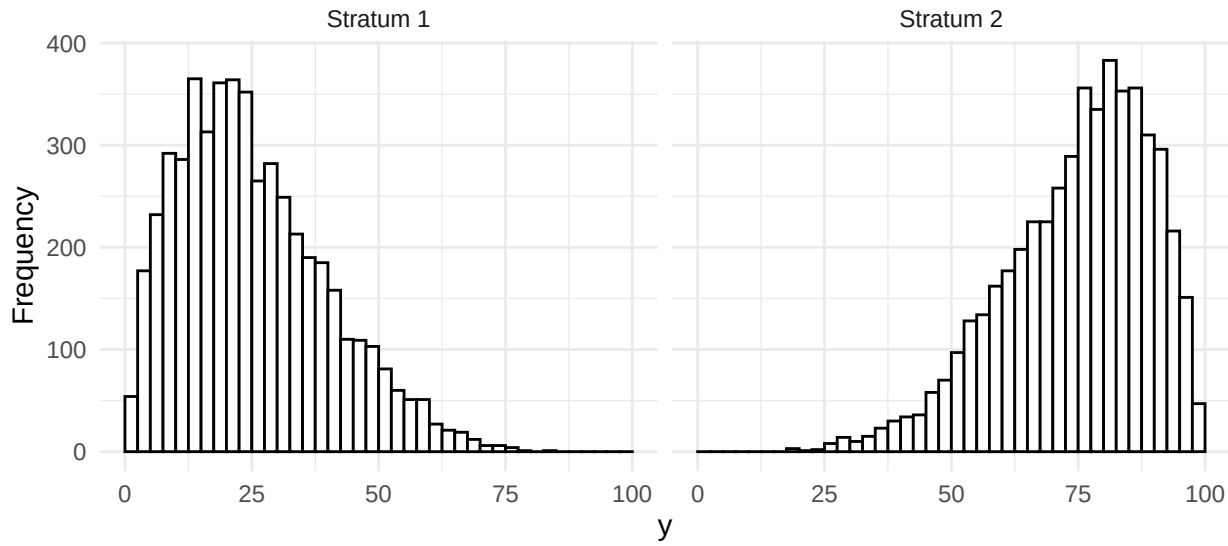
Simulation Results

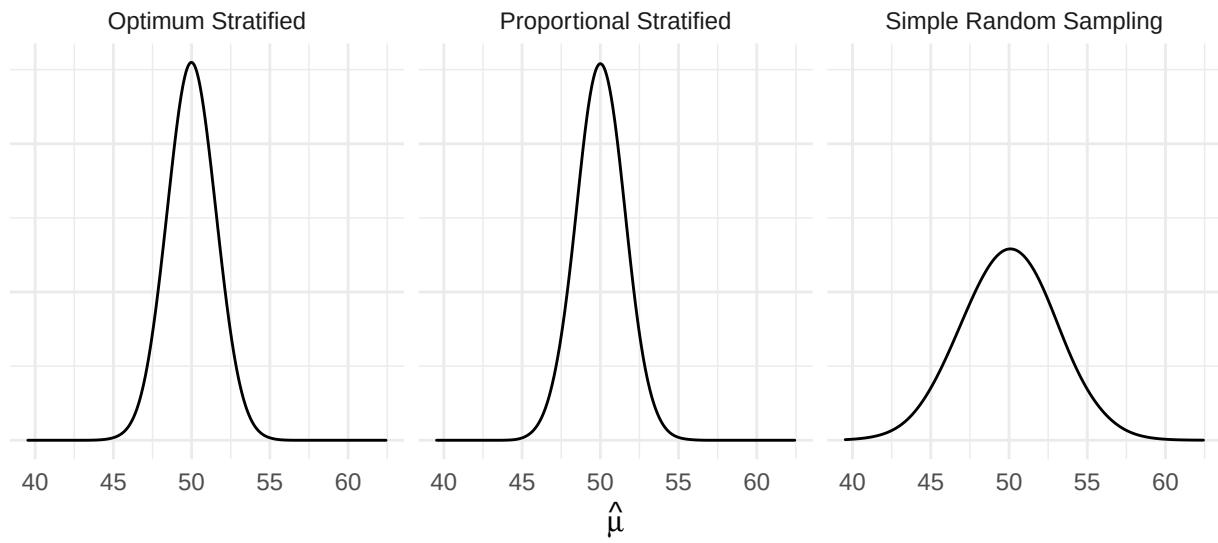
Strata have *same* means, *different* variance.



design	V	SE	B
Optimum Stratified	2.61	1.62	3.24
Proportional Stratified	3.90	1.97	3.94
Simple Random Sampling	3.91	1.98	3.96

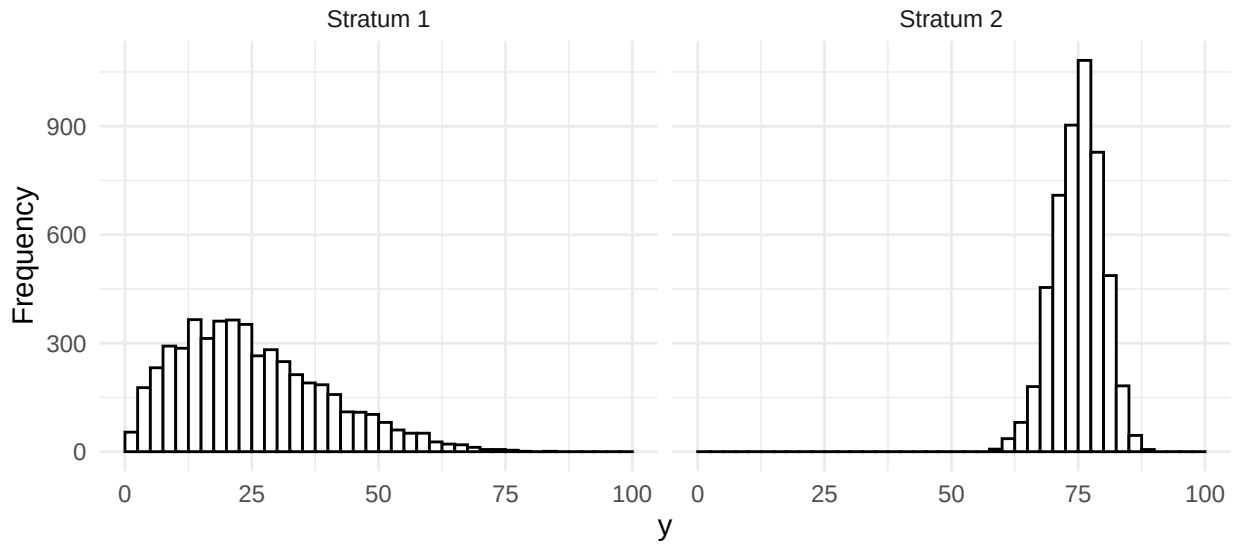
Strata have *different* means, *same* variance.

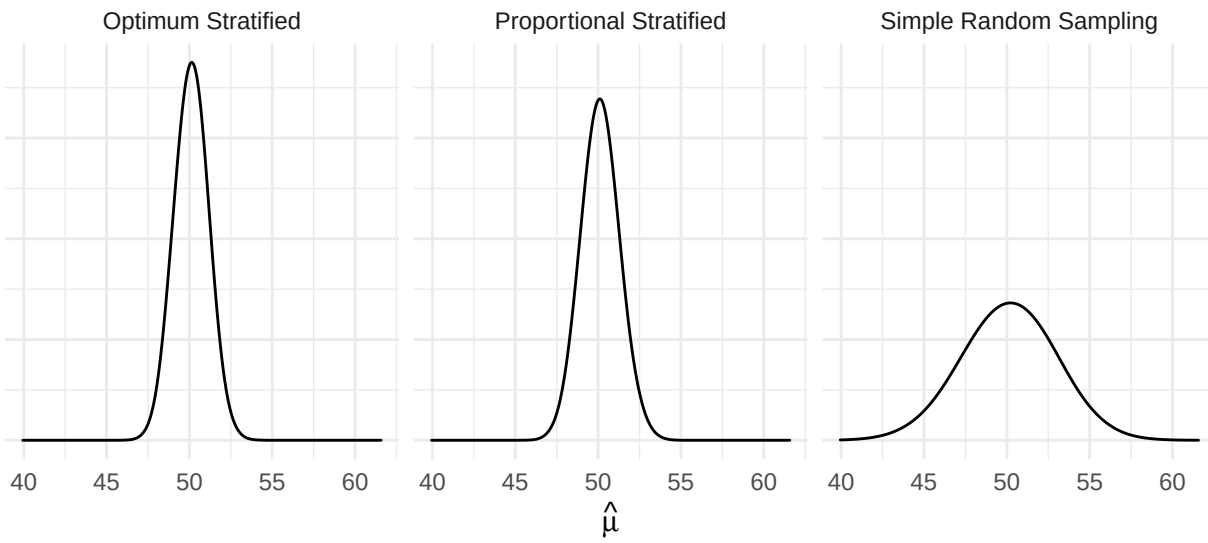




design	V	SE	B
Optimum Stratified	2.09	1.45	2.90
Proportional Stratified	2.11	1.45	2.90
Simple Random Sampling	8.05	2.84	5.68

Strata have *different* means, *different* variance.





design	V	SE	B
Optimum Stratified	0.93	0.96	1.92
Proportional Stratified	1.18	1.09	2.18
Simple Random Sampling	7.18	2.68	5.36

Design Effect

The **design effect** of a complex sampling design is defined as

$$D = \frac{V_{\mathrm{C}}}{V_{\mathrm{SRS}}}$$

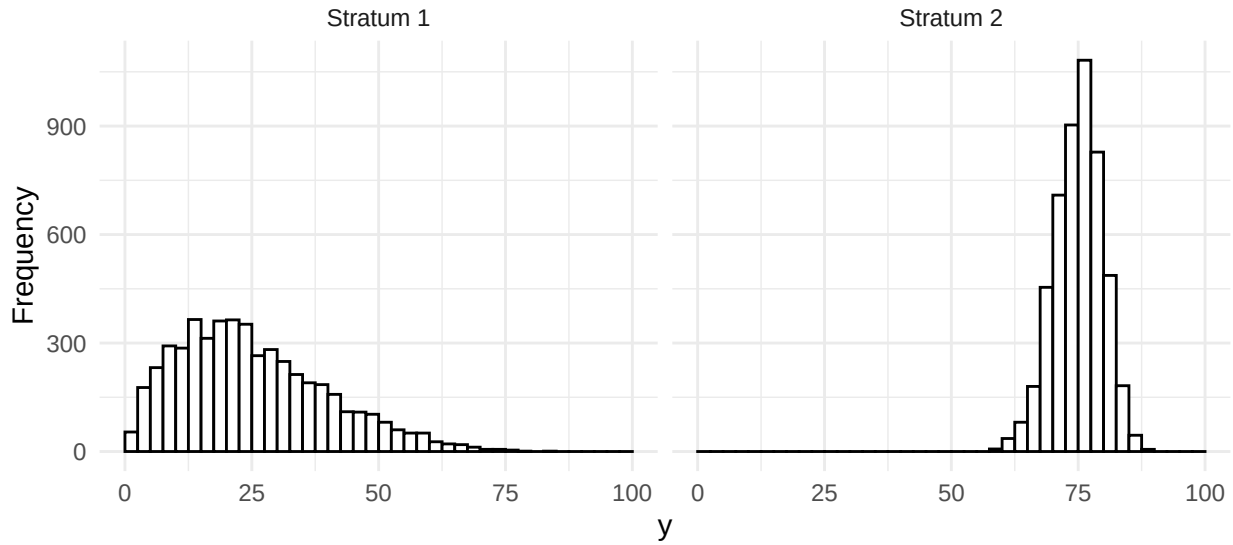
where V_C and V_{SRS} are the variances of the estimators for a paramater under a complex sampling design (e.g., stratified random sampling) and simple random sampling, respectively. Clearly

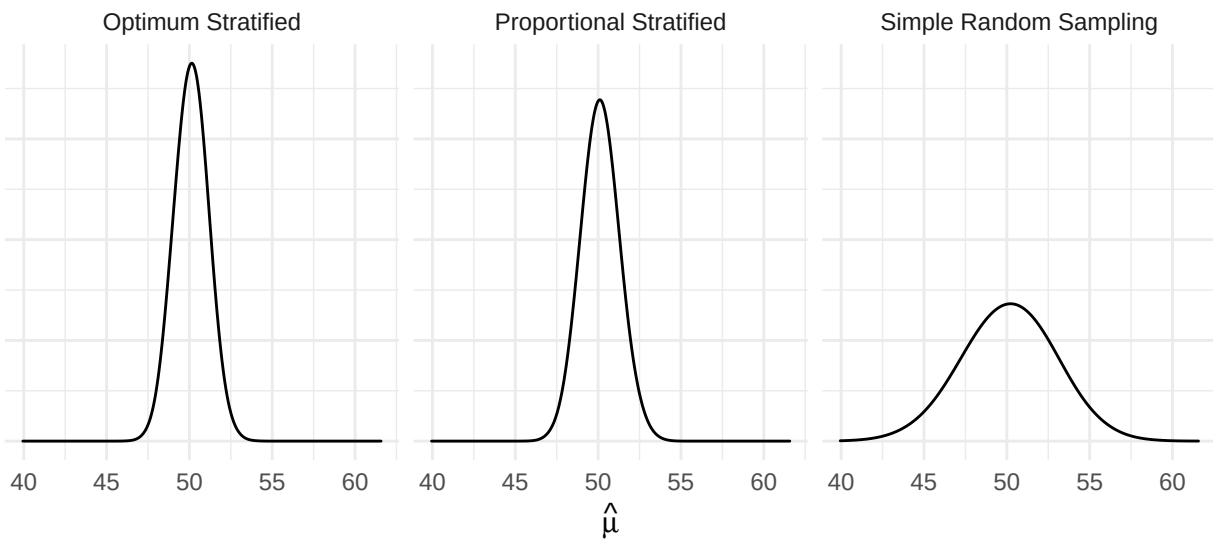
$$\begin{aligned} D < 1 &\Leftrightarrow V_{\mathrm{C}} < V_{\mathrm{SRS}} \quad \text{(i.e., complex is better)}, \\ D = 1 &\Leftrightarrow V_{\mathrm{C}} = V_{\mathrm{SRS}} \quad \text{(i.e., same)}, \\ D > 1 &\Leftrightarrow V_{\mathrm{C}} > V_{\mathrm{SRS}} \quad \text{(i.e., SRS is better)}. \end{aligned}$$

The design effect can be computed a couple of ways.

1. Hypothetical analysis (using equations or simulation).
2. Estimated from a sample drawn using the complex sampling design.

Example: Recall the simulation with the strata with unequal means and variances.





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Optimum Stratified	0.93	0.96	1.92
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From this simulation we have that $V_{\text{Opt}} \approx 0.93$, $V_{\text{Prop}} \approx 1.18$, and $V_{\text{SRS}} \approx 7.18$. What are the design effects of the two stratified random sampling designs?

Effective Sample Size

The **effective sample size** of a complex sampling design is defined as

$$\text{ESS} = \frac{n}{D},$$

and is interpreted as the sample size that a simple random sampling design would need for an estimator to have the same variance as that based on a complex sampling design.

Example: The sample size used in the simulation above is $n = 100$. What is the *effective sample size* of the two stratified random sampling designs?